

Advanced Applied Econometrics:

Dynamic labor supply - A simple life-cycle framework

Summer 2026

In order to understand the core principles of solving finite-horizon models like the one in KW97 via backward induction, this problem set asks you to solve a simplified setup with limited heterogeneity and a short time horizon.

Single-agent framework: We consider a single-agent life-cycle model of labor supply and consumption decisions, a stripped-down version of the Blundell et al. (2016) model, tracing out behavior from labor-market entry onwards. The goal is to illustrate the dynamic incentives shaping labor supply and consumption over a finite horizon.

Time horizon and agent/worker:

- Start with a small time horizon: three periods (e.g. years since labor-market entry), equivalent to using age. We index periods by $t = 0, 1, 2$.
- Every period the worker (notation abstracts from the individual index i) maximizes lifetime utility (the baseline is deterministic; the expectation becomes relevant only once wage shocks are added in Problem 6) given current characteristics and the economic environment: age t , work experience e_t , and a two-dimensional unobserved work-disutility term $\theta = (\theta_P, \theta_F)$ over part-time (P) and full-time (F) employment. We denote the vector of state variables at age t by X_t .
- *Convention:* we order $\theta = (\theta_P, \theta_F)$; Blundell et al. (2016), eq. (1), use the reverse order (θ_F, θ_P) . Keep this in mind when cross-referencing the paper.

Assumptions on utility:

- Utility is *inter-temporally separable* (additively separable across periods t), but *within* a period it is *non-separable in consumption and leisure*. Instantaneous utility depends on current consumption c_t , labor supply l_t , and work preferences θ . The worker's flow utility at age t is (cf. Blundell et al. (2016), eq. (1)):

$$u(c_t, l_t; \theta) = \frac{c_t^\mu}{\mu} \exp[U(l_t, \theta)]$$

- Labor supply l_t is a discrete choice with three alternatives: out of work (O), part-time (P), and full-time (F). We use “out of work” / non-employment (the paper's O) rather than “unemployment”, since no job search is modeled. The term $U(\cdot)$ shifts the marginal utility of consumption with the labor-supply choice and preferences θ ; it is normalized to 0 out of work.
- $\mu < 0$ is the curvature parameter governing risk aversion and the elasticity of intertemporal substitution. Because $\mu < 0$, flow utility c_t^μ / μ is **negative**, so a **positive** $U(\cdot)$ scales it

by $\exp[U] > 1$ and pushes utility *further* below zero. Hence working lowers the utility of consumption, $\theta_P, \theta_F > 0$ are **disutilities** of work, and consumption and labor supply are complements (cf. Blundell et al. (2016), p.1734).

- With $\theta_F > \theta_P > 0$, full-time work is more costly than part-time.

$$U(l_t, \theta) = \begin{cases} 0, & \text{if } l_t = O, \\ \theta_P, & \text{if } l_t = P, \\ \theta_F, & \text{if } l_t = F. \end{cases}$$

Decision problem: At every age t , the worker chooses labor supply l_t and consumption c_t to maximize expected lifetime utility, given the current state X_t and the future consequences of current choices:

$$V_t(X_t) = \max_{\{c_\tau, l_\tau\}_{\tau=t, \dots, T}} E \left[\sum_{\tau=t}^T \beta^{\tau-t} u(c_\tau, l_\tau; \theta) \mid X_t \right] \quad \text{s.t. the budget constraint.}$$

- The expectation $E[\cdot]$ is taken over future random events conditional on X_t ; β is the discount factor; V_t is the optimal value of discounted present and future utility.
- Here consumption c_t is pinned down by the labor-supply choice l_t (see the budget constraint below), so the choice reduces to l_t .
- NOTE: this baseline setup contains no stochastic elements; it is purely deterministic.

Budget constraint / available consumption:

- For this problem set we abstract from the full tax-transfer system, assets, and savings in Blundell et al. (2016). Available consumption c_t is therefore pinned down directly by current labor earnings.
- Earnings (and hence consumption) are measured **per period**, as a function of the per-period wage rate w_t and a per-period hours choice $h_t \in \{0, 24, 38\}$ (corresponding to O, P, F). We deliberately leave time units abstract: all flows are per model period.

$$c_t = w_t h_t \quad \text{for } l_t \in \{P, F\}.$$

- Out of work there are no earnings. To stand in for the transfer system (out-of-work benefits) and to keep flow utility well-defined for $\mu < 0$ (where $c \rightarrow 0$ sends $c^\mu/\mu \rightarrow -\infty$), we impose a **consumption floor** $\bar{c} > 0$ when out of work:

$$c_t = \bar{c} \quad \text{for } l_t = O.$$

- This floor is the only deviation of the consumption side from the paper, forced by dropping assets and $T(\cdot)$; it plays the role of the income floor implied by income support in Blundell et al. (2016).

Human capital and earnings dynamics: The worker's hourly wage w_t is a deterministic function of accumulated work experience e_t and a constant productivity intercept. The log-log specification gives curvature in the wage-experience profile, consistent with the empirical evidence.

Experience accumulation is a function of the previous period's labor-supply choice, $g(l_{t-1})$. In principle one normalizes full-time accumulation to $g(l_{t-1} = F) = 1$ and estimates the relative value of part-time experience $g(l_{t-1} = P)$ from the data. For illustration we instead fix integer accumulation directly, $g(l_{t-1} = P) = 1$ and $g(l_{t-1} = F) = 2$, so that experience is always integer-valued and the state space stays small. This both inverts the paper's normalization ($g_s(F) = 1$) and raises the *relative* part-time value to $g(P)/g(F) = 0.5$, well above the paper's estimate of 0.10-0.15; it is a purely computational device, so do not read structural meaning into these values (see the calibration note below).

$$\begin{aligned}\ln(w_t) &= \gamma_0 + \gamma_1 \ln(e_t + 1) \\ e_t &= e_{t-1} + g(l_{t-1}) \\ e_0 &= 0 \quad (\text{no work experience at labor-market entry})\end{aligned}$$

Calibration of the model:

- $\mu = -0.5$, giving relative risk aversion $1 - \mu = 1.5$ (the paper uses $\mu = -0.56$, i.e. RRA = 1.56).
- $\beta = 0.98$ (as in the paper).
- Wage equation: intercept $\gamma_0 = 2$, experience slope $\gamma_1 = 0.2$.
- Experience accumulation: $g(l_{t-1} = P) = 1$, $g(l_{t-1} = F) = 2$ (integer device; see the note above on the inverted normalization and the inflated part-time ratio relative to the paper's 0.10-0.15).
- Part-time hours $h_t = 24$ (raised from 18) so that part-time earnings clear the out-of-work floor $\bar{c}$ already at $e = 0$ and part-time remains a relevant alternative at low experience; with $h_t = 18$, part-time consumption at labor-market entry would fall below the floor and, combined with the disutility θ_P , part-time would never be attractive early on.
- Disutility of work: $\theta_P = 0.1$, $\theta_F = 0.34$ (both **positive**, $\theta_F > \theta_P$, so work lowers utility and full-time costs more; consistent with the sign logic of $U(\cdot)$ above). θ_F is set close to the static full-time / non-employment indifference threshold so that the human-capital channel is pivotal: a myopic agent and a forward-looking agent need not make the same choice.
- Out-of-work consumption floor: $\bar{c} = 150$ (out-of-work-benefits stand-in).

Problem 1: General questions

1. What are the structural parameters in the wage and utility specification of this framework? Discuss their role in the model and how they influence behavior.
2. **Experience accumulation:**
 - Discuss the process governing the endogenous accumulation of human capital / work experience. What is captured by $g(l_{t-1})$?
 - Review Blundell et al. (2016), p.1727f: How does their specification $e_t = e_{t-1}(1 - \delta) + g(l_{t-1})$ differ from this setup? What is the role of the structural parameter δ ? What human-capital / wage dynamics can be captured with this setup?
3. **Budget constraint:** Review the more comprehensive budget constraint in Blundell et al. (2016), p.1726. Discuss the implications of the more comprehensive specification in the paper for the worker's decision making. And more generally, how does the budget constraint differ from the static setting in terms of dynamic incentives?

Problem 2: Static setting and implications of dynamic incentives

1. Assume again a static setting (or a myopic agent with $\beta = 0$), where the worker only considers the first period $t = 0$ in their labor supply decision. Write down and describe the decision problem of the worker. What determines the optimal choice?
 - Period 0: experience stock is zero.
 - Derive utility for each choice alternative.
2. Compute the utility of each choice alternative and the optimal labor supply choice for the worker in the first period $t = 0$.
3. Now assume that the worker is forward-looking. Discuss the dynamic incentives influencing the labor supply choice at any point in time if the worker considers some finite time horizon. Would you expect different behavioral implications for the worker in period $t = 0$?

Problem 3: State space

In this problem we will define a useful representation of the state space. Obtaining the solution to the model will require us to compute all relevant states (combinations of state variables) that the model can reach, and derive the objects determining optimal agent behavior (value functions) at each of these states.

1. **State-space:** What are the state variables of the model? How many states are there? Count only *reachable* states, i.e. combinations of state-variable values that can actually occur given the initial condition $e_0 = 0$ and the experience-accumulation technology $g(\cdot)$. Write down the state space as a matrix with dimensions *number of states* by *number of state variables*.
2. **State-space container and indexer:** In discrete choice dynamic programming problems like this one, it is helpful to create a container of the state space, and an associated indexer that allows the easy look-up of the values of all state variables at any state with index i , and vice-versa. Write a function that generates these objects:
 - `states` (np.array): A 2d array of shape (num_states, num_state_variables) with all possible states.
 - `indexer` (np.array): A 2d array of shape (num_periods, max_possible_exp) with the index of each state in the states array.

Verify the indexer returns the expected state index for a given input of state-variable values.

Problem 4: Solving the model with backwards induction: step-by-step

In discrete choice dynamic programming problems like this one, the agent makes choices based on choice-specific value functions given the current state X_t . Solving the model will thus entail the calculation of the choice-specific value functions at each state of the state space. Recall that in every period t the worker maximizes expected discounted lifetime utility as defined above.

1. Using the principles introduced in the IO lecture on dynamic discrete choice (in an infinite horizon setting), derive the recursive Bellman expression for the value function $V(X_t, \theta)$ and the choice-specific value functions $V^l(X_t, \theta)$, $l \in \{O, P, F\}$. (Cp. expressions on slide 37 of lecture 4.) Discuss its components and parameters. What are the implications of the deterministic setting for these expressions?
2. In a finite-horizon setting, we can derive the value functions at each state by backwards induction. Assume that in the final period continuation values are zero. Compute the choice-specific value functions at each state and report the optimal labor supply choice for the worker in the final period 2. Use the state-space container and indexer generated above to generate inputs for the utility / value function.
3. Next, the value functions computed in the previous exercise can be used as continuation values for the derivation of the value functions in the second-to-last period ($t = 1$). Complete the backwards induction and the choice-specific value functions at each state of periods 0 and 1. What would be the optimal labor supply choice at each state? Describe your solution procedure.
4. Having derived the model solution, i.e. the objects governing behavior at each possible state of the model, discuss the optimal behavior of the agent in each period. In particular: is full-time work ever *statically* optimal, i.e. optimal on flow-utility grounds alone, at any experience level? Show why (not), and relate this to the multiplicative form of $u(c_t, l_t; \theta)$ with $c_t = w_t h_t$.

Problem 5: Solve and simulate the model

You have found the solution of the model when you have calculated the choice-specific value functions at each state.

1. **Basic simulation:** Using the model solution, simulate a dataset with the decision path, i.e. the optimal choice in each period, for a single agent in the 3-period setting. Your simulation output should be a matrix or a matrix-like object with columns “Period”, “Choice”, “Choice-specific value function”. In addition, simulate the decision path of a *myopic* agent ($\beta = 0$) over the same three periods and compare the two paths. What explains the difference?
2. **Larger time horizon:** Your program is general and scalable if you can use the exact same code in order to solve the worker’s decision problem for 30 periods.

Problem 6: Extensions of the framework

1. **Individual heterogeneity:** Discuss how one could introduce relevant individual observed and unobserved heterogeneity into this simple framework. Discuss one option each. What are the implications for the state space?
2. **Introduction of stochastic elements:** Suppose we introduce a stochastic element into the wage process $\ln(w_t) = \gamma_0 + \gamma_1 \ln(e_t + 1) + \varepsilon_t$, where ε_t , the stochastic fluctuation of wages, is assumed to be i.i.d. normally distributed with mean zero, $\varepsilon_t \sim N(0, \sigma_\varepsilon^2)$. As in Problem 4.1, write down the Bellman equation for the value function and describe what changes compared to the deterministic setting. What do we need to assume about the

timing of the shock, i.e. whether ε_t is realized before or after the labor-supply choice l_t is made? Assume the shock is observed by the agent and realized before making the labor supply choice. Write down the Bellman equation. Is the shock a state variable? Does it enter the state-space container required to solve the model?